

EcoTrack | AI Prompt Documentation

Project: Personal Carbon Footprint Calculator

AI tool: ChatGPT (AI-assisted HTML, CSS, and JavaScript development). The prompts below document the design instructions and refinement criteria for this project. They are presented as a reproducible prompt plan, not as a verbatim chat transcript.

1. Sustainability problem and scope

Design a beginner-friendly web calculator that estimates annual personal greenhouse gas emissions from car travel, flights, household electricity, natural gas, and diet. Explain that estimates are illustrative and avoid implying scientific precision.

2. Core software implementation

Create a self-contained responsive single-page website using HTML, CSS, and vanilla JavaScript. Include sliders for weekly driving, annual round-trip flights, monthly electricity and gas, plus a diet selector. Update annual CO₂e totals immediately when inputs change.

3. Transparent calculations

Calculate emissions in tonnes of CO₂e per year. Use clearly stated illustrative factors and convert weekly and monthly usage to annual usage. Include transport, home energy, and food breakdowns; guard against zero totals and negative results.

4. Visual and accessibility refinement

Use a calm green-and-cream design, readable type, mobile-responsive two-column layout, explicit input labels, accessible chart descriptions, and simple horizontal bars that show relative category contributions.

5. Actionable sustainability recommendations

Show up to three recommendations based on actual user inputs. Add a numerical scenario estimating savings from driving 20% less and using 10% less electricity. Do not promise that hypothetical savings will occur.

6. User experience and testing

Add a reset control and a download-results feature without external libraries. Check slider updates, dropdown updates, zero-use cases, scenario math, responsive layout, and output download. Include a GitHub Pages publishing README.

Submission note

Upload the project files to GitHub, enable GitHub Pages, test the public URL, and submit that functioning link with this PDF. If your instructor requires exact prompts entered during development, replace this planned prompt documentation with your actual conversation transcript.